Radius of Validity for Partial Taylor Series Approximations of a Sinusoid

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Problem Setup

This document develops an analytic expression for the radius of validy of a taylor series approximation to a sinusoid as a function of the number of terms in the approximation.

Let $T_N\{f\}(u)$ denote the *N* term Taylor Series expansion for $f(u)$ about $u = 0$:

$$T_N\{f\}(u) = \sum_{n=0}^{N} \frac{f^{(n)}(0)}{n!} u^n$$

$$T_N\{\sin(u)\}(u) = \sum_{n=0}^{N} \frac{(-1)^n}{(2n+1)!} u^{2n+1}$$

Let the error between the the *N* term Taylor Series expansion for $f(u)$ and $f(u)$ itself be denoted:

$$E_N\{f\}(u) = T_N\{f\}(u) - f(u)$$

We want to $\rho\,(N)$ such that $T_N\{\sin(u)\}(u)$ will satisfactorily approximate $\sin(u)$ inside the interval $(-\rho\,(N), \rho\,(N))$. A satisfactory approximation is defined as an approximation $T_N\{f\}(u)$ for which $E_N\{f\} \leq \max(f(u))\ \forall\, u \in (-\rho\,(N), \rho\,(N))$.

Solution

Note that due to the Lagrange Remainder Theorem:

$$|E_N\{f\}(u)| \leq \frac{M}{(n+1)!} |u|^{n+1} \quad s.t. \quad M \geq |f^{(n+1)}(u)|$$

Since $\max\left(\frac{\mathrm{d}^k}{\mathrm{dx}^k} \sin(u)\right) = 1\ \forall\, k$, we may choose $M = 1$

Then it follows that the worst case error for the *N*term Taylor Series approximation $T_N\{\sin(u)\}(u)$is:

$$E_N\{\sin(u)\}(u) = \frac{u^{n+1}}{(n+1)!}$$

Now, for some $n = N$we wish to find the value of *u* for which the error $E_N\{\sin(u)\}(u)$ overshadows the $\sin(u)$ function itself. That is, we wish to find:

$$\rho\,(N) = u \quad s.t. \quad E_N\{\sin(u)\}(u) = \max(\sin(u))$$

Since $\max(\sin(u)) = 1$, we wish to find:

$$\rho\,(N) = u \quad s.t.\ \frac{u^{N+1}}{(N+1)!} = 1$$

Simplifying:

$$\rho\,(N) = \sqrt[N+1]{(N+1)!}$$

If we assume $\rho\,(N)$ is approximately linear, we can derive an approximation for it's slope and intercept.

Noting that we have two points:

$$\rho\,(N) = \sqrt[N+1]{(N+1)!}$$

$$\rho\,(N+1) = \sqrt[N+2]{(N+2)!}$$

Using these two points to define a line and manipulating algebraically

$$m = \frac{\sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!}}{N+1-N},\ b = N\,\frac{\sqrt[N+1]{(N+1)!} - \sqrt[N+2]{(N+2)!}}{N-(N+1)} - \sqrt[N+1]{(N+1)!}$$

$$m = \frac{\sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!}}{1},\ b = N\,\frac{\sqrt[N+1]{(N+1)!} - \sqrt[N+2]{(N+2)!}}{-1} - \sqrt[N+1]{(N+1)!}$$

$$m = \sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!},\ b = N\,\frac{\sqrt[N+1]{(N+1)!} - \sqrt[N+2]{(N+2)!}}{-1} - \sqrt[N+1]{(N+1)!}$$

$$m = \sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!},\ b = N\left(\sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!}\right) - \sqrt[N+1]{(N+1)!}$$

$$m = \sqrt[N+2]{(N+2)!} - \sqrt[N+1]{(N+1)!},\ b = Nm - \sqrt[N+1]{(N+1)!}$$

$$m = \rho\,(N+1) - \rho(N),\ \ b = N\,m - \rho\,(N)$$

As we will see below, the linear approximation is useful, but an much more accurate formula using Sterling's approximation is derived:

Starting again with:

$$\rho\,(N) = \sqrt[N+1]{(N+1)!}$$

And noting that according to Sterling's approximation:

$$k! \sim \sqrt{2\pi\, k}\left(\frac{k}{\mathrm{e}}\right)^{k}$$

And applying Sterling's approximation to $\rho\,(N)$:

$$\rho\,(N) \sim \left(\sqrt{2\pi\,(N+1)}\left(\frac{N+1}{\mathrm{e}}\right)^{N+1}\right)^{\frac{1}{N+1}}$$

Simplifying:

$$\rho\,(N) \sim (2\pi\,(N+1))^{\frac{1}{2(N+1)}}\left(\frac{N+1}{\mathrm{e}}\right)$$

Thus the radius of validity of an *N* term partial Taylor Series expansion where $N >> 1$ of a sinusoid is given:

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$$\rho\,(N) \sim \sqrt[2(N+1)]{2\pi\,(N+1)}\left(\frac{N+1}{\mathrm{e}}\right)$$

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The above formula allows evaluating $\rho\,(N)$ for large *N* without explicitly computing a factorial. We will see below that it has error well below the linear approximation for all but the values immediately adjacent to the point about which the linear approximation is calculated.

Testing the Result

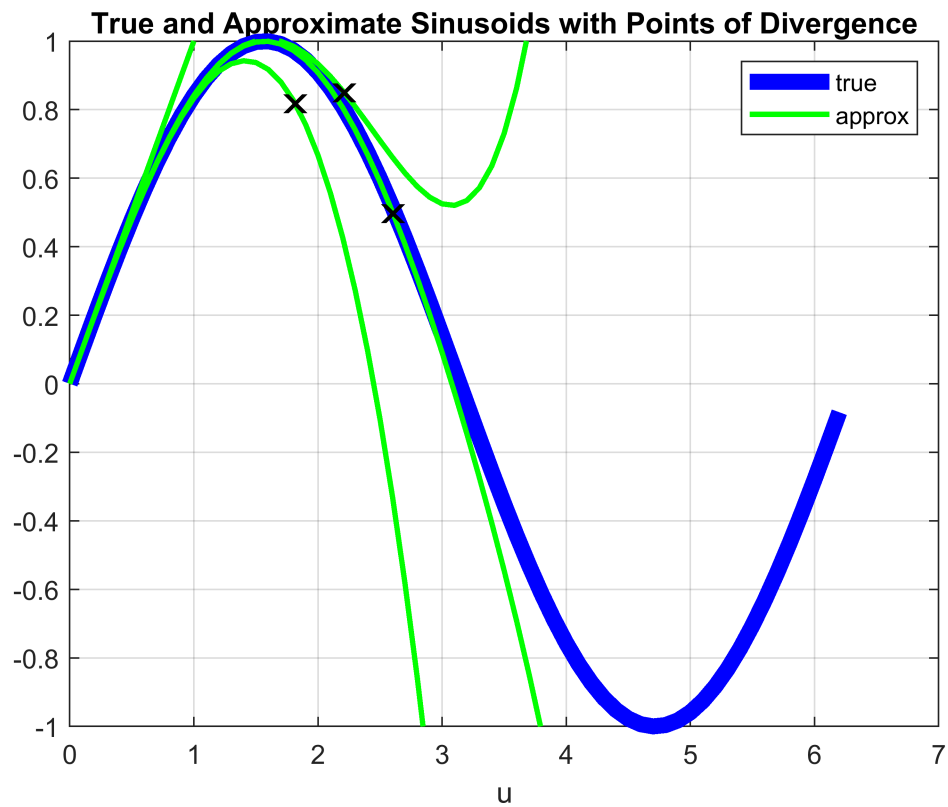
```
du = 0.1;
u = 0:du:2*pi;
f_true = sin(u);

num_terms_list = 0:70;
f_approx = zeros(length(u), length(num_terms_list));
rho = zeros(1, length(num_terms_list));
approx_sin_rho = zeros(1, length(num_terms_list));

for num_terms_idx = 1:length(num_terms_list)
    n = num_terms_list(num_terms_idx);
    f_approx(:,num_terms_idx) = ComputeTaylorSine(u, n);
    rho(num_terms_idx) = ComputeRho(n);
    approx_sin_rho(num_terms_idx) = ...
        ComputeTaylorSine(rho(num_terms_idx), n);
end

figure;
max_terms = 5;
plot(u, f_true, 'color', 'b', 'linewidth', 6);
hold on;
plot(u, f_approx(:, 2:max_terms), 'color', 'g', 'linewidth', 2);
plot(rho(2:max_terms), approx_sin_rho(2:max_terms), 'kx', 'linewidth', 2, 'markersize', 10);
ylim([-1 1]);
```

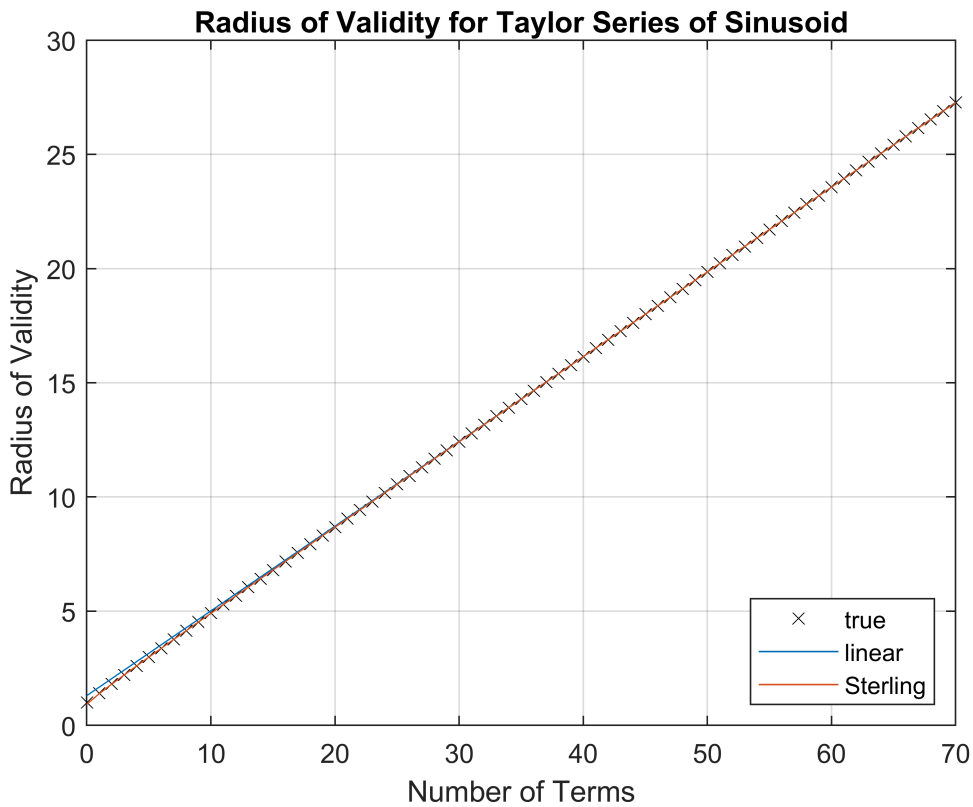
```
xlabel('u');
title('True and Approximate Sinusoids with Points of Divergence')
legend('true', 'approx');
grid on;
```



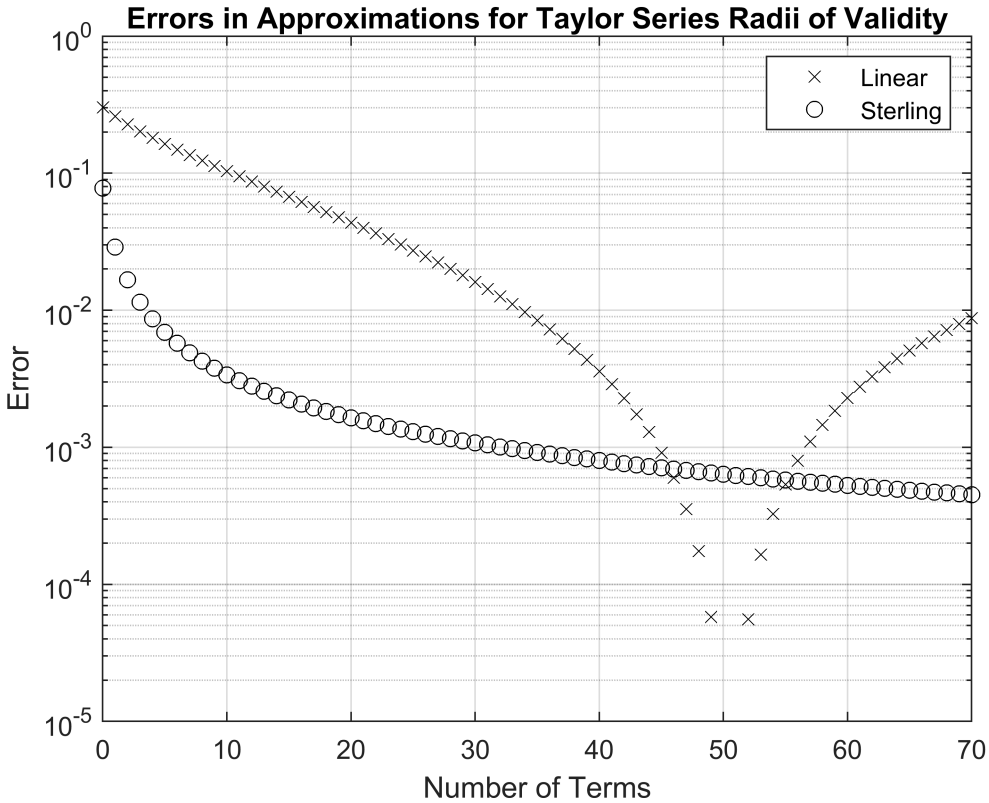

```
m_rho = ComputeM(50); % large number chosen arbitrarily for center of linear approx.
b_rho = ComputeB(50);
rho_linear_approx = m_rho*num_terms_list + b_rho;
rho_large_n_approx = ComputeRhoSterlingFormula(num_terms_list);

error_linear_approx = abs(rho - rho_linear_approx);
error_large_n_approx = abs(rho - rho_large_n_approx);


figure;
plot(num_terms_list, rho, 'kx');
hold on;
plot(num_terms_list, rho_linear_approx);
plot(num_terms_list, rho_large_n_approx);
grid on;
title('Radius of Validity for Taylor Series of Sinusoid');
legend({'true', 'linear', 'Sterling'}, 'location', 'southeast');
xlabel('Number of Terms');
ylabel('Radius of Validity');
```




```
figure;
semilogy(num_terms_list, error_linear_approx, 'kx');
hold on;
semilogy(num_terms_list, error_large_n_approx, 'ko');
grid on;
xlabel('Number of Terms');
ylabel('Error');
title('Errors in Approximations for Taylor Series Radii of Validity');
legend('Linear', 'Sterling');
```




```
function approx_sine = ComputeTaylorSine(u, num_terms)
approx_sine = zeros(size(u));
for n = 0:num_terms-1
    a = ((-1).^n)/factorial(2*n+1);
    approx_sine = approx_sine + a*u.^(2*n+1);
end
end

function rho = ComputeRho(n)
rho = (factorial(n+1)).^(1/(n+1));
end

function m = ComputeM(n)
m = ComputeRho(n+1) - ComputeRho(n);
end

function b = ComputeB(n)
b = ComputeRho(n) - n*ComputeM(n);
end

function rho = ComputeRhoSterlingFormula(n)
rho = (2*pi*(n+1)).^(1./(2*(n+1))).*(n+1)./exp(1);
end
```